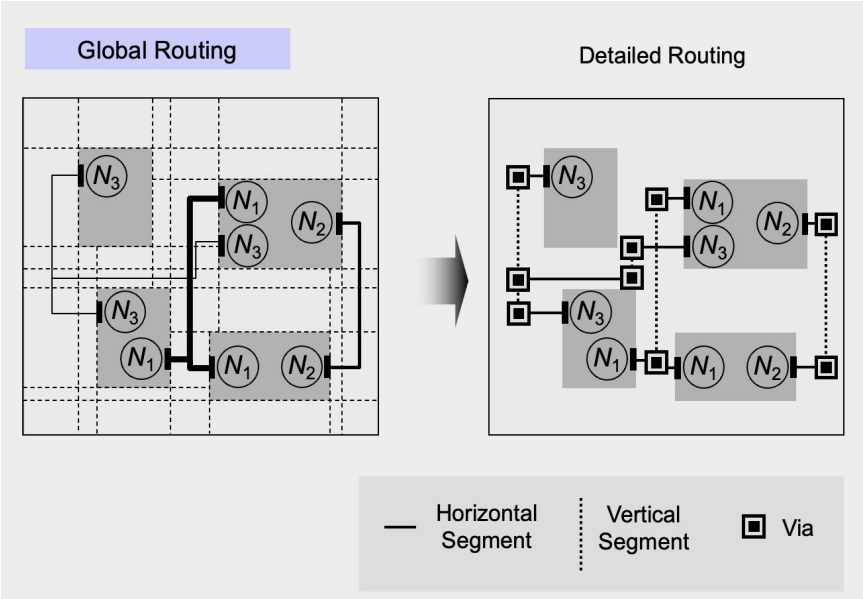
CSE460: VLSI Design

Lecture 15+16: VLSI Physical Design
Routing Algorithm

Introduction

- Forming physical connection between pins. First, approx. path (**Global Routing**), then, placing metal line segment (**Detailed Routing**)
- After **placement** phase, the exact **locations** of **circuit blocks** and **pins** are determined.
- Space not occupied by the blocks are used for routing and are called as **routing regions**.
- Two kinds of approaches to solve global routing problem: **Sequential** & **Concurrent** Approach.



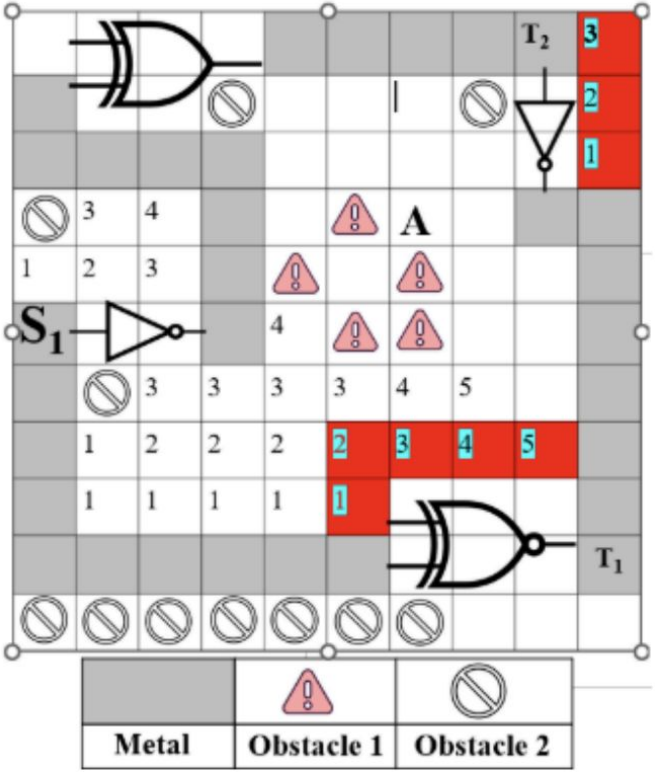
Routing Approaches

Sequential Approach:

- In this approach, as the name suggests, pins are routed **one by one**.
- ***Order is important***: once a region has been routed it may block other nets which are yet to be routed.
- Eg.
  i. **Maze routing** algorithms
  ii. Line-probe algorithms
  iii. Shortest path based algorithms

Concurrent Approach:

- This approach **avoids the ordering problem** by considering routing of all the nets simultaneously.
- Computationally sophisticated as memory requirement is **very high** due to parallel operation.



Here, multiple square cells are occupied by a single gate. Hence, each source or target pin is occupied by a single square cell as shown in the picture.

Lee's Maze Routing algorithms

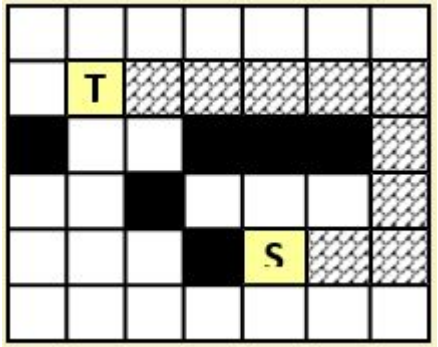
- The layout surface is assumed to be made up of a rectangular array of **grid cells**.
- Each ***Grid cell*** represents a square cell where ***one*** wire can cross.
- Objective is to find out the ***shortest*** path (**sequence of grid cells**) for <u>connecting two points</u> (Source, **S** to Target, **T**) .
- When using cells, a wire can either ***cross*** or ***bend***.

cross ***Bend***

- Some of the grid cells act as **obstacles (Black Cells)**.

  -Blocks that are placed on the surface.

  -Some nets that are already laid out.



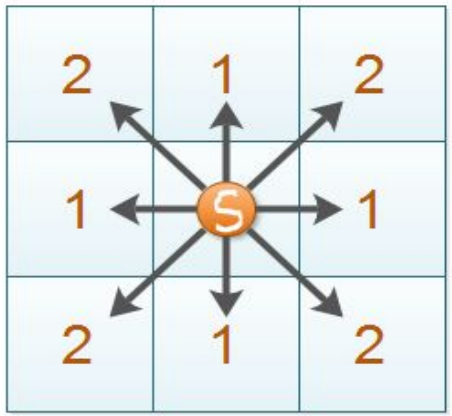
Lee's Algorithm : *Expansion*

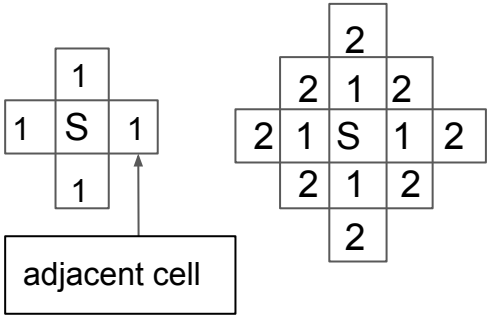
Step 1: Wave propagation -

- Iterative process
- Starts at the most adjacent cell of the Source.
- Need to find all new cells/grid that are reachable at ***pathlength 1***(i.e, all paths that are just 1 unit in total length(just 1 cell)).
- Using the ***pathlength 1*** cells, all new cells which are reachable at ***pathlength 2*** can be found.
- Process is repeated until the target, **T** is reached.
- During i^{th} iteration, <u>non-blocking grid cells</u> at Manhattan distance of i from grid cell **S** are all <u>labeled with</u> **i**

Note: Manhattan distance is the number of strides (horizontal and/or vertical) required to reach a cell.

Manhattan Distance



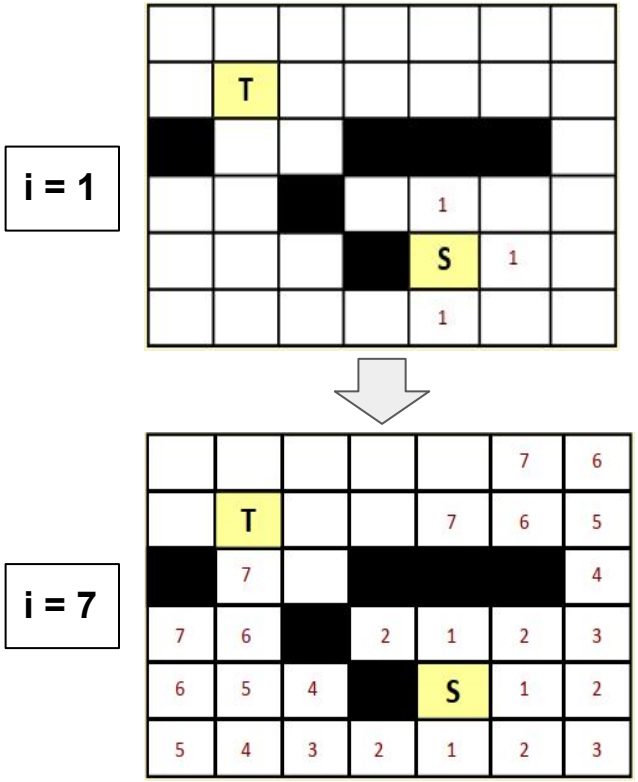


Lee's Algorithm

Step 1: Wave propagation (contd)-

- Labeling continues until the target grid cell **T** is marked in iteration L (i.e. when i=L0).
- L0 is the length of the shortest path.

- The process fails if:
  - T is not reached and no new grid cells can be labeled during step i.
  - T is not reached and i equals M, some upper bound on the path length.

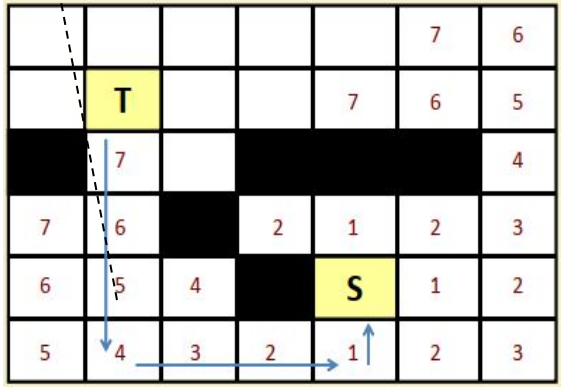


Lee's Algorithm

direction change means going to the 4 in the right from 5 instead of 4 in it's below position

Step 2: Retrace-

- Systematically **backtrack** from the target cell **T** back towards the source cell **S**.
- If T was reached during step i, then at least one grid cell adjacent to it will be labeled i-1, and so on.
- By tracing the numbered cells in descending order, we can reach S following the shortest path.

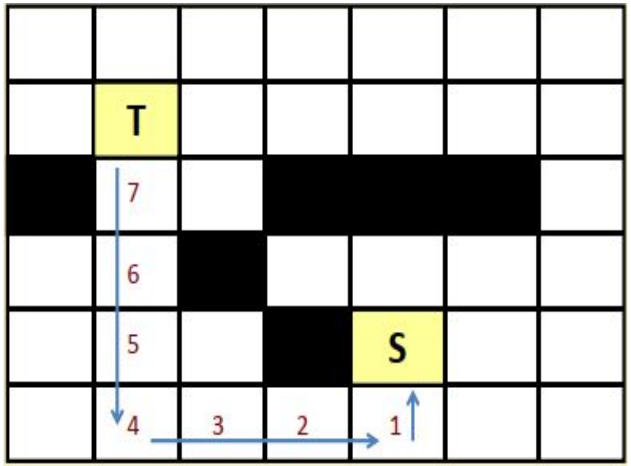


- In practice, the **rule of thumb** is not to change the direction of retrace unless one has to do so.
- **Minimizes** number of **bends to reduce system complexity.**

Lee’s Algorithm

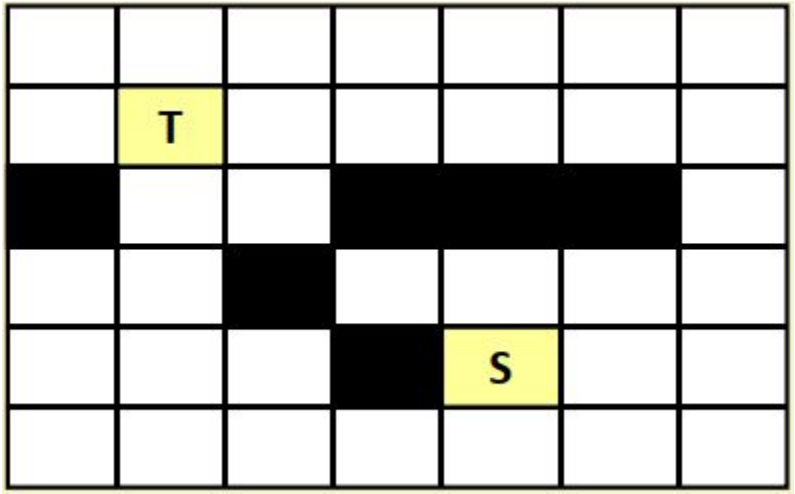
Step 3: Label clearance-

- All labeled cells except those corresponding to the path just found are **cleared**.
- Cells along the path are marked as **obstacles**.
- Search complexity is as involved as the wave propagation step itself.



Example 1

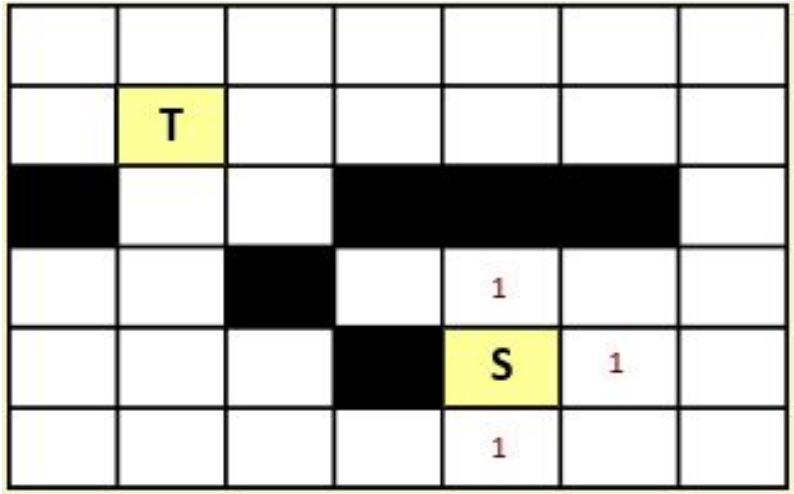
Initial routing problem:



Example 1

Step 1: Wave propagation

i = 1



Example 1

Step 1: Wave propagation

i = 2

	T					
			2	1	2	
				S	1	2
			2	1	2	

just proceed one manhattan distance horizontally/vertically from cells labelled as ‘1’

Example 1

Step 1: Wave propagation

i = 3

	T					
			2	1	2	3
				S	1	2
		3	2	1	2	3

Example 1

Step 1: Wave propagation

	T					
						4
			2	1	2	3
		4		S	1	2
	4	3	2	1	2	3

i = 4

	T					5
						4
			2	1	2	3
	5	4		S	1	2
5	4	3	2	1	2	3

i = 5

Example 1

Step 1: Wave propagation

						6
	T				6	5
						4
	6		2	1	2	3
6	5	4		S	1	2
5	4	3	2	1	2	3

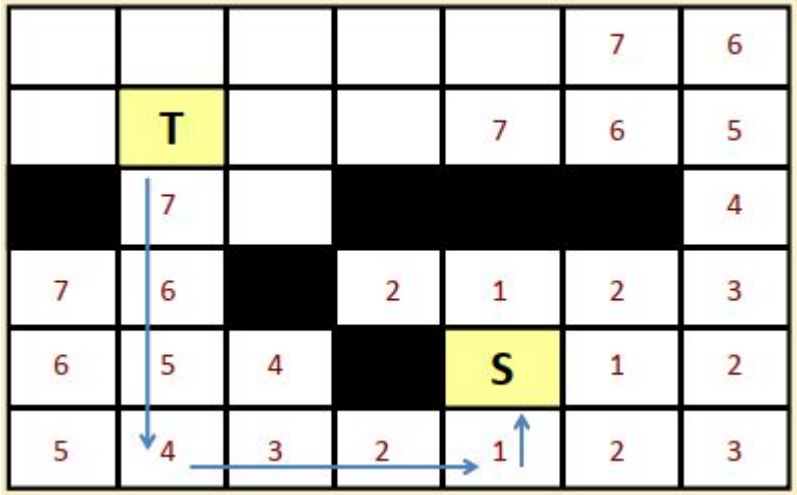
i = 6

					7	6
	T			7	6	5
	7					4
7	6		2	1	2	3
6	5	4		S	1	2
5	4	3	2	1	2	3

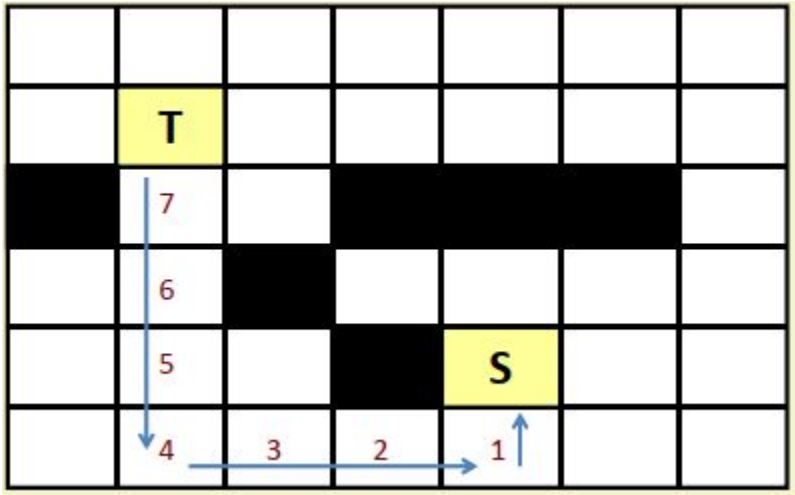
i = 7

Example 1

Step 2: Backtrace

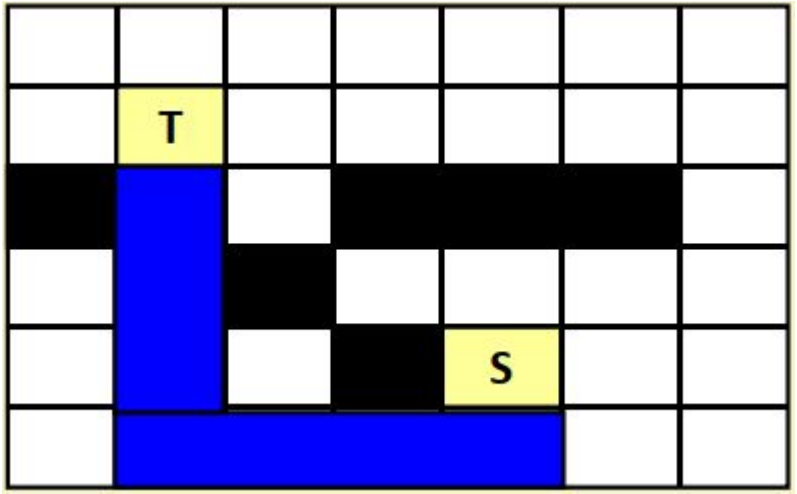


Step 3: Clear



Example 1

Final output (Mark):



- Marked region is now blocked.
- New wave cannot propagate through this merged region unless the same S has another target T.
- If the source has another target T, then whole marked (blue) region will be considered as Source, S which we will explore in Example 2.

Multi-Terminal Nets

- Step 1: Propagate wave from the source s to the closet target.
  - One of the terminals (**A**; out of 5) of the net is treated as source, and the rest as targets.
  - A wave is propagated from the source until one of the targets is reached.
- Step 2: <u>All the cells in the marked path are next labeled as source cells</u>,(total path from A-B is considered as 1 source) and the remaining unconnected terminals as targets.
- Step 3: Propagate wave from ALL s cells to the other cells.
- Step 4: Continue until all cells are reached.
- Step 5: Apply heuristics to further reduce the tree cost

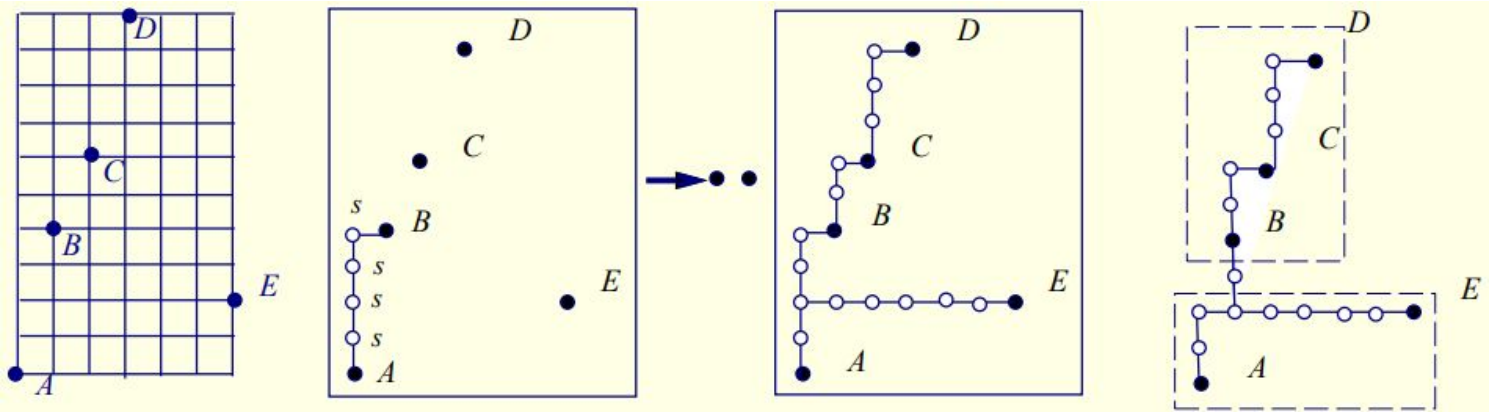
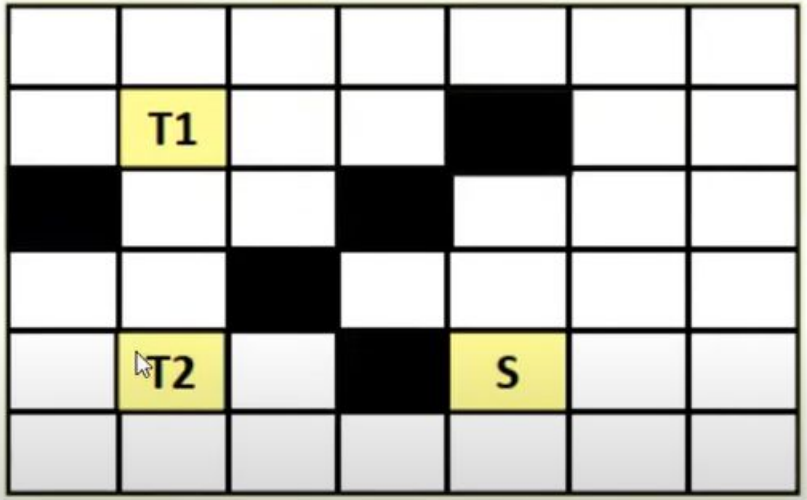


Fig: (a) A as source; B-E as targets (b) A-B as source; C-E as Targets. (c) all paths from source to target were found (d) optimized path from A-B (*low bend*)

Example 2

goal: same S must be connected to T1 and T2

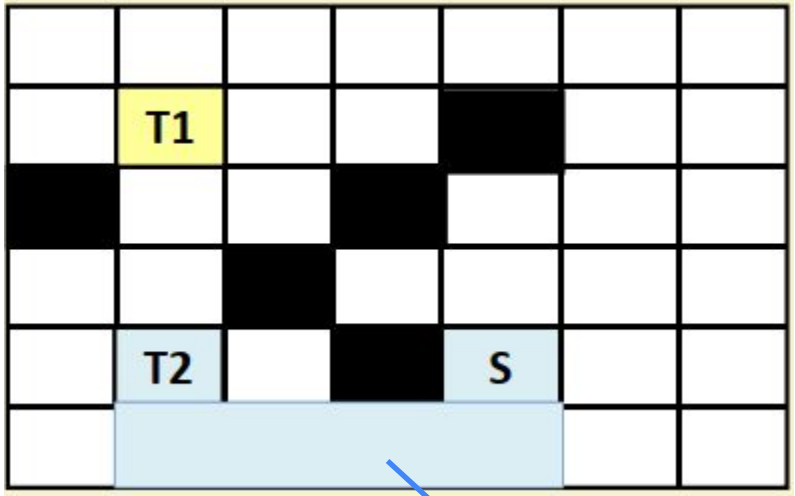


Extension of Lee's algorithm:

- One of the terminals (out of 3 here) of the net is treated as source, and the rest as targets.
- A wave is propagated from the source until one of the targets is reached.
- <u>All the cells in the determined path are next labeled as source cells</u>, and the remaining unconnected terminals as targets.
- Process continues.

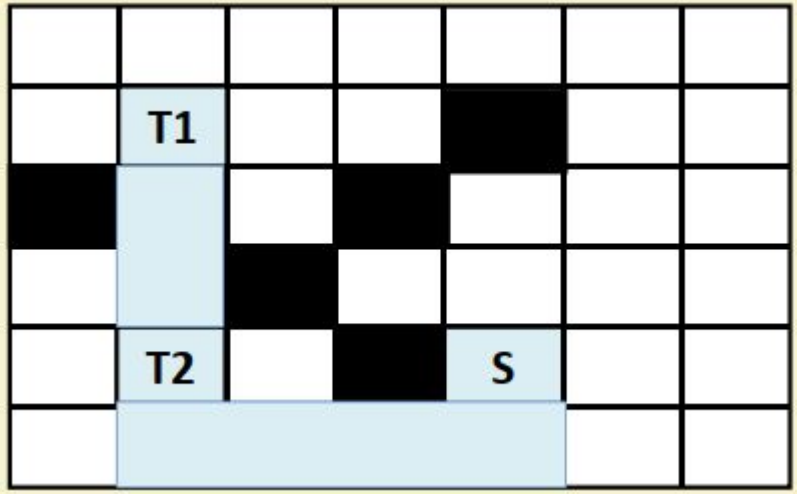
Example 2

Step 1 to 3:



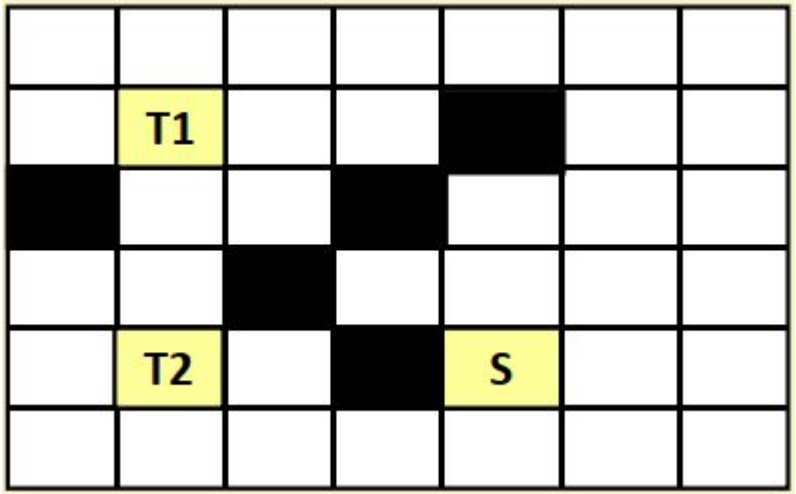
this whole metal bar now acts as the source

Step 1 to 3 (Again):

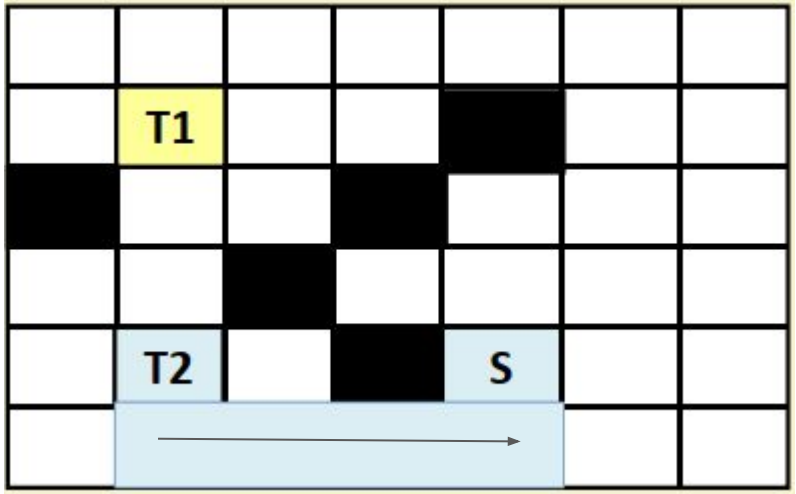


Ref. [1]

Example 2: Extension of Lee's algorithm[Multi point nets/ Target]

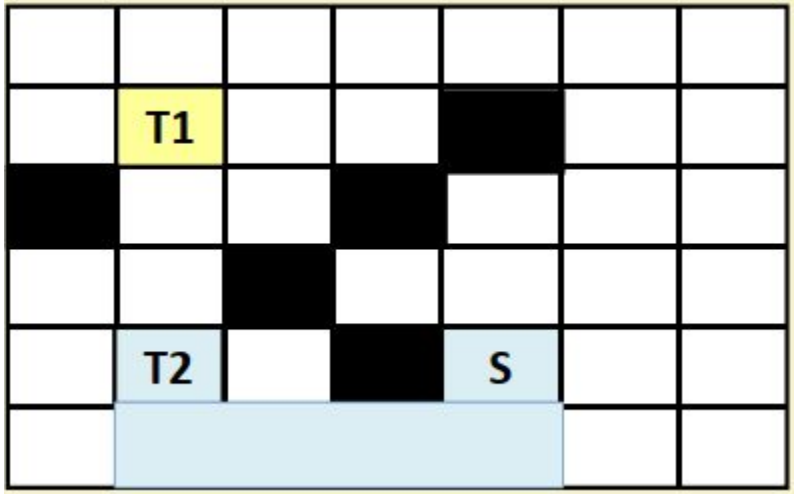


(a) Initial state

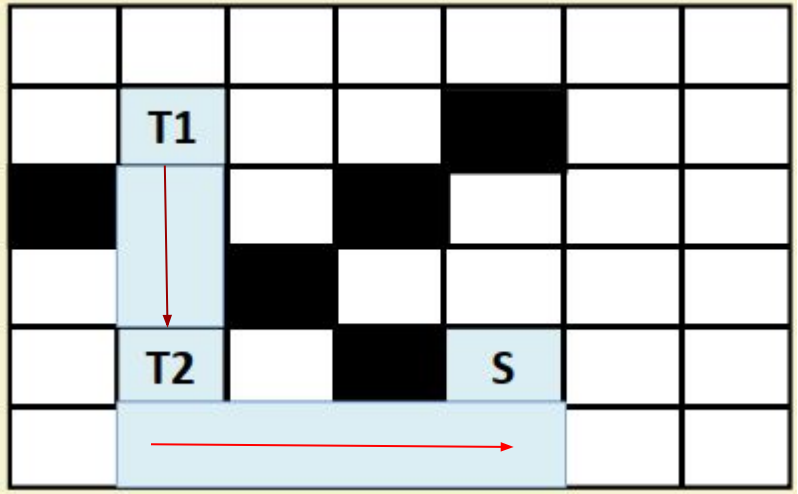


(b) Finding Path-1; From S to T2

Example 2 : [Multi point nets/ Target]

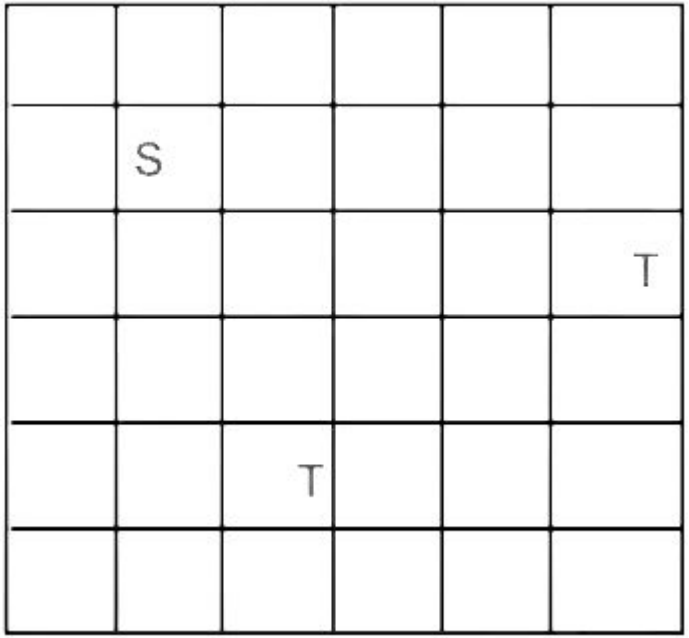


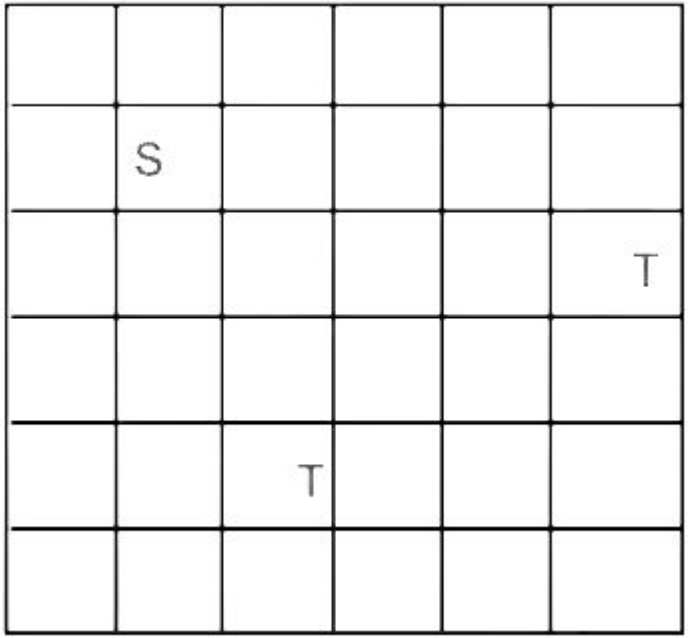
(c) S-T2 path as a source, ***S1*** for T1



(d) Finding Path-2; From ***S1*** to T2

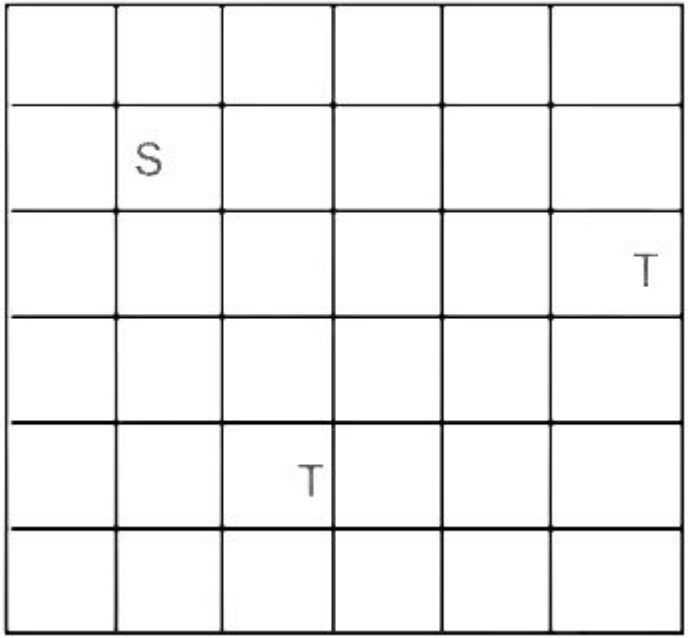
Example 3: CW/HW

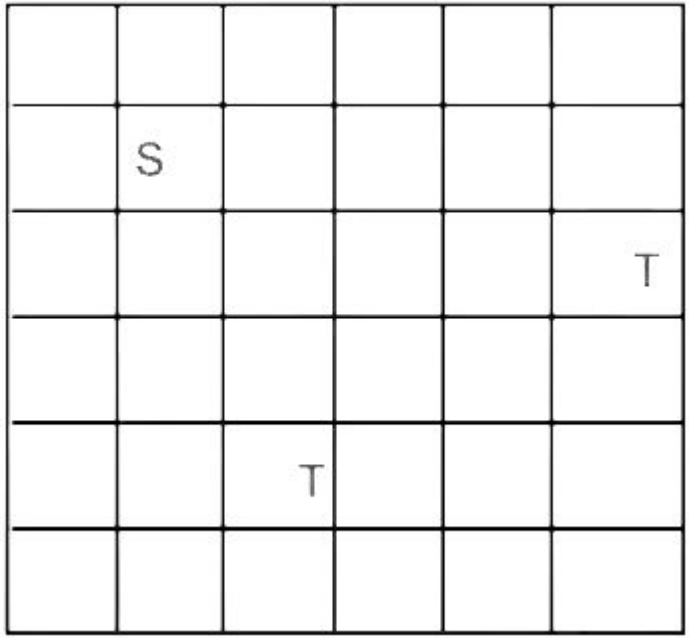




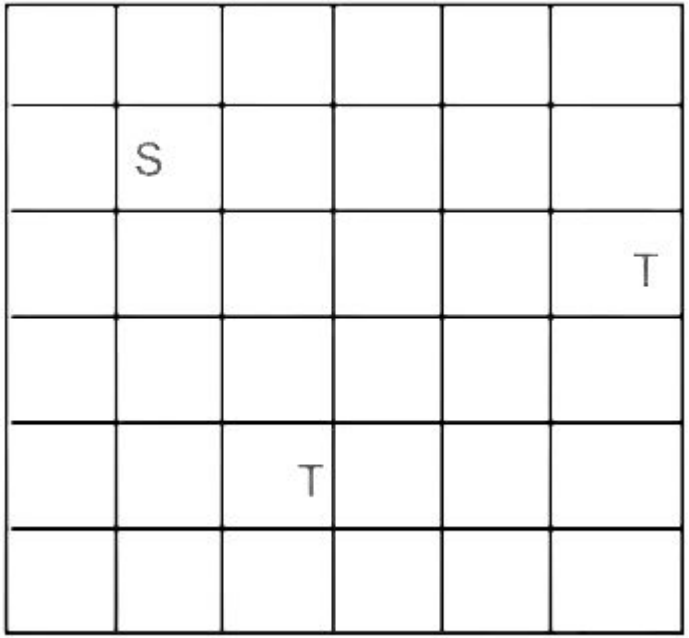
- Find the shortest path for each Target.
- Calculate the memory usage.

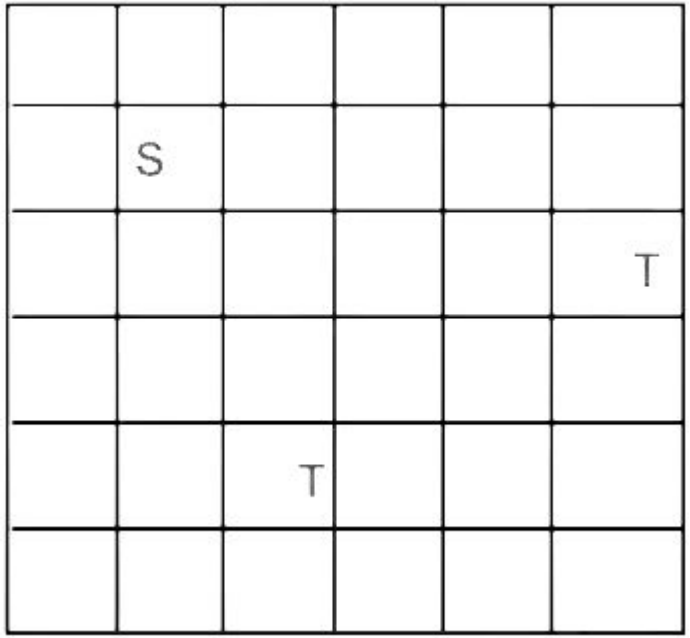
Example 3: CW/HW





if we take the green colored path, we would need to fill additional 3 grids with metal to reach the second target instead of 2 grids while taking the blue path as shown in the next slide. Hence, it’s not the optimum result


S
T
T


S
T
T

Memory Requirement

- Each cell needs to store a number between 1 and L, where L is some bound on the maximum path length.
- For ***M x N*** grid, ***L*** can be at most ***M+N-1***.
- **Two things** yet to be denoted: empty cell/obstacle.
- So, $n = ceil(log_2(L+2))$ bits per cell
- Total memory = (***M x N x n***) bits

			6
			5
1	2	3	4

Here, 3 x 4 grid, 3 bits(000=blank; 111 may mean obstacle; 6==110,5=101 etc) are required per cell
Total memory = 36 bits

Memory Requirement

Examples:

1. **2000 x 2000** grid
   n = ceil($\log_2$ (4001))= 12
   Memory required = 2000 x 2000 x 12 bits = **6** Mbytes

2. **3000 x 3000** grid
   n = $\log_2$ (6001) = 13
   Memory required = 3000 x 3000 x 13 bits = **14.6** Mbytes

Note: For memory requirement calculations, the maximum path length (**L**) is always considered to be (**M+N-1**) if the sequence is a series of natural numbers (1,2,3,4,5,....). Actual path length can be lower or equal to this value. If we use a repeating sequence, L is total number of unique entity.

Memory Optimization

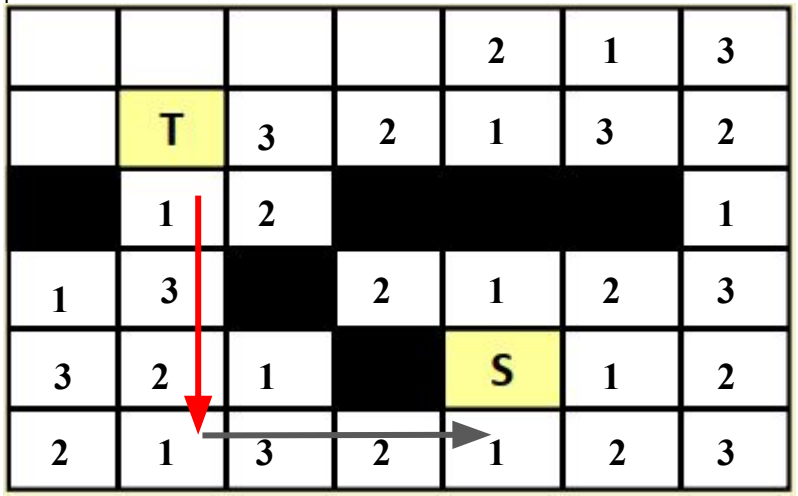
Akers's Observations (1967)

- Adjacent labels for k are either k − 1 or k + 1. Need a labeling scheme such that each label has its preceding label different from its succeeding label.

WAY 1

- Instead of using the sequence 1,2,3,4,5,….. for numbering the cells, the sequence 1,2,3,1,2,3,… is used.(***L=3***)
- For a cell, labels of predecessors and successors are different. So tracing back is easy.
- ***ceil(log$_2$(3+2))*** = ***3*** bits per cell.

1.5 Mbytes for 2000 x 2000 grid
(< 6Mbytes in prev slide)



Memory Optimization [*Minimum Cost*]

WAY 2

- Use the sequence 0,0,1,1,0,0,1,1,….., (***L*=2**)
- 0-> First 0; 1-> First 1.
- 0-> Second 0; 1-> Second 1.
- Predecessors and successors are again different.
  0, $\boxed{0}$,1
- ***ceil(log$_2$(2+2))*** **= 2** bits per cell.

2000 x 2000 x 2 bits / 8 =1.0 Mbytes for 2000 x 2000 grid

to represent 4 states: blank, obstacle, 0 ,1

					1	0
	T			1	0	0
	1					1
1	0		0	0	0	1
0	0	1		S	0	0
0	1	1	0	0	0	1

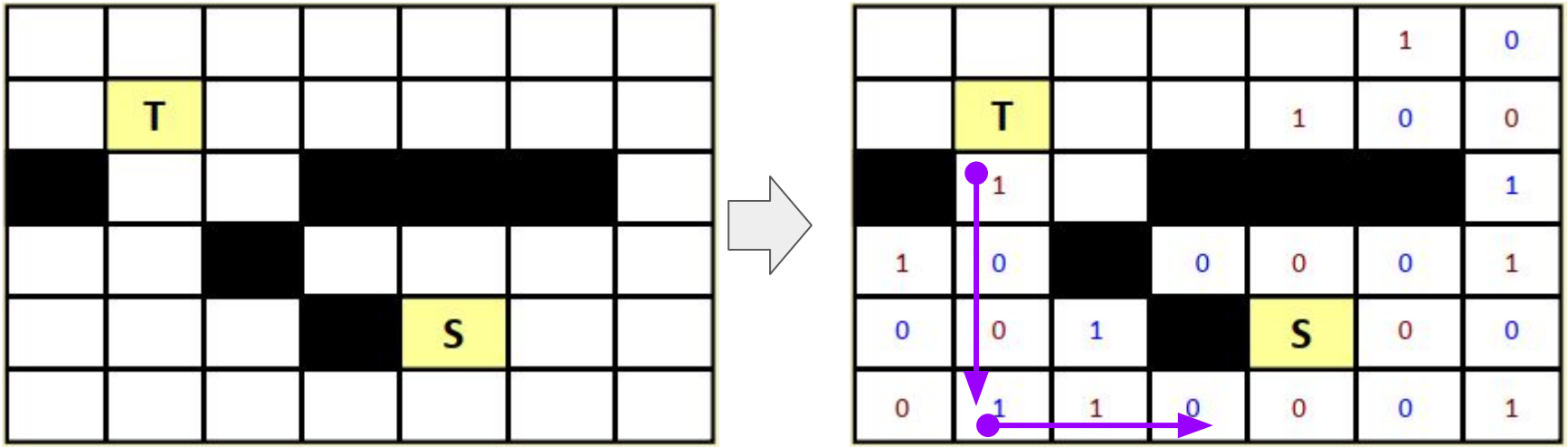
Fig: Minimum Bend Path

Memory Requirements - Summary

- For memory requirement calculations, the critical path length (L) is always considered to be (M+N-1) if the sequence is a series of natural numbers (1,2,3,4,5,....). Actual path length can be lower or equal to this value.
- If we use a repeating sequence, L is the total number of unique entities. For example, if we use 1,2,3,1,2,3... repeating sequence, L = 3.
- If the sequence is 0,1,0,1...., L = 2. So, L does not depend on the actual path length for memory calculation.

If you are asked to calculate the memory used by a definite path, that's a different question. Generally, ***memory requirements*** denote the maximum memory required to run the algorithm.

Example 4: Using 00110011..... Sequence



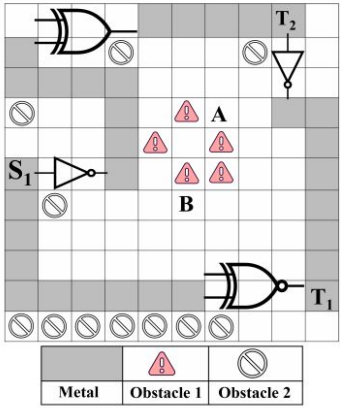
Label: 0011001

Retrace

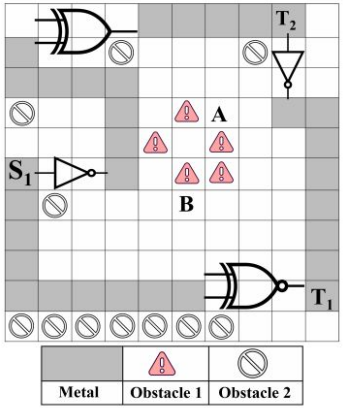
For question (a) you are given a certain chip planning task. You have four blocks with the following widths and heights:
Block A: w = 1, h = 5 or w = 5, h = 1
Block B: w = 1, h = 3 or w = 3, h = 1
Block C: w = 2, h = 1 or w = 1, h = 2
Block D: w = 3, h = 3 or w = 1, h = 9
For questions (b-e) refer to the figures at the bottom of the question. You can see the grid layout of a system with a source (S1) and two targets – T1 and T2. The obstacles and free cells are shown.

(a)	**Find** and draw a floor plan with the minimum total area enclosed by the global bounding box using 10x10 space.	**[3]**
(b)	**Determine** the minimum Manhattan distance from A to B (You don't have to show in grid. Just write the answer).	**[1]**
(c)	Using the Lee's Maze Algorithm, **find** the shortest path from **S1** to **T1** and **T2** *allowing minimum bends.* **Answer this question on this and the subsequent grids on the bottom of the question paper. A and B are not obstacles.**	**[4]**
(d)	**Find** the total memory usage of the given grid if the sequence is a series of natural numbers. If the maximum memory requirement of the provided grid is 28 bytes, calculate the maximum number of bits that each cell can hold.	**[2+3]**
(e)	**Find** the number of consecutive and unique numbers (starting from 0) that can be used for wave propagation in the Lee's Maze algorithm, for the given memory constraint in (d). What should be a full sequence for wave propagation? **[2]**	

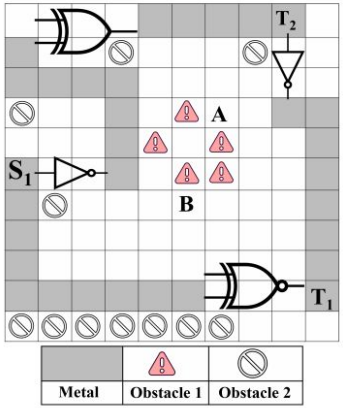
Grid 1



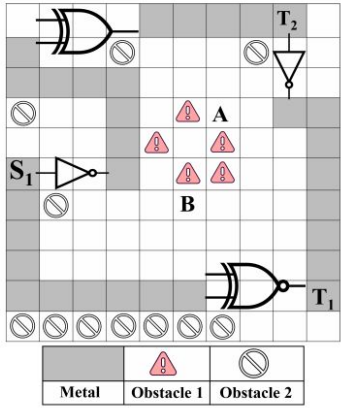
Grid 2



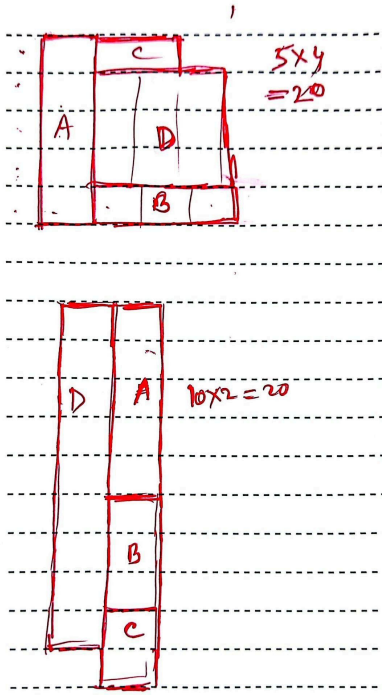
Grid 3



Grid 4



a.

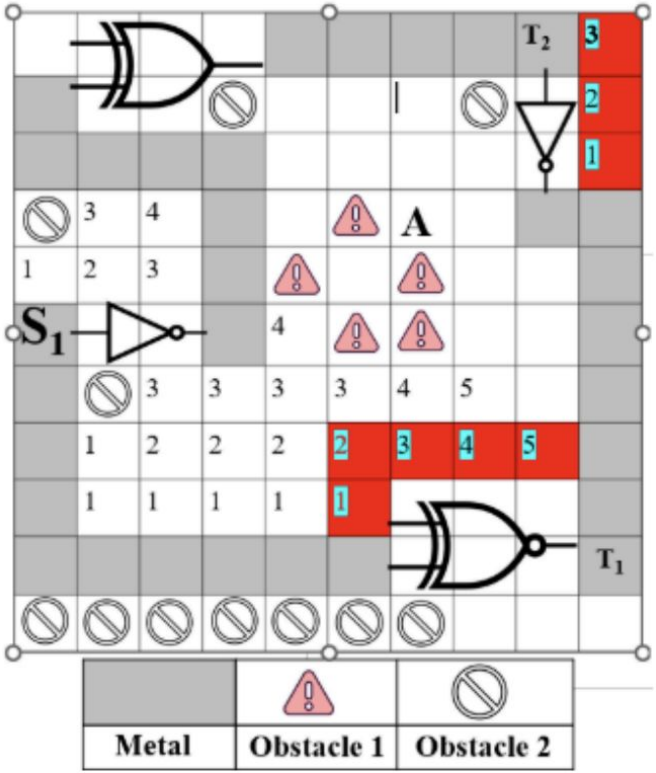


(b) 6 Manhattan Distance (treat the metal layer as a single node hence till B(not before B)= total = 6

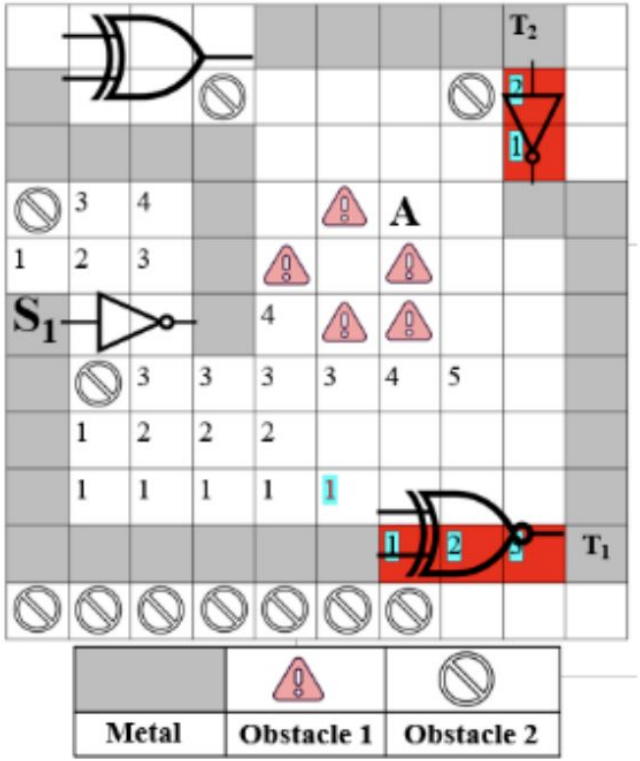
(the cell containing the destination,B must also be considered)

c.

considering gates as obstacles:



by not considering gates as obstacles:



(d)

$M = 10, N = 11$

$\therefore L = M + N - 1$

$= 20$

$n = \text{ceil}(\log_2(20+2))$

$= \text{ceil}(4.45)$

$= 5$

$\therefore$ Total Memory $= M \times N \times n$

$= 10 \times 11 \times 5$

$= 550$ bits

$= 68.75$ bytes

(e) Memory $= 28$ bytes

$= 224$ bits

$\therefore n = \frac{224}{M \times N} = 2.04$

$= 2$

$\therefore L = 2^n - 2 = 2$

Sequence $= 0, 1, 0, 1, 0, 1, \ldots$

References

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3. http://users.eecs.northwestern.edu/~haizhou/357/lec6.pdf